Ryan Henderson

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EDUCATION

University of Delaware, Honors College — Newark, DE
Bachelor of Electrical Engineering — May 2027
GPA: 3.89; Dean's List: Fall 2023 - Spring 2026

SKILLS

Programming/HDL: Python, C, C#, MATLAB, Verilog; EDA: KiCad, LTspice, Vivado, OpenROAD, Yosys, Icarus Verilog;
Lab/Test: Oscilloscopes, Spectrum/Logic Analyzers; Fabrication: E-beam PVD, diffusion, wet etching, metallization

WORK EXPERIENCE

Siemens Healthineers — Tarrytown, NY
Electrical Engineering Intern — June 2026 - August 2026

- Developed a C# application interfacing with a USB-to-I²C bridge to automate EEPROM serialization of PCAs, implementing read-back verification, page-aware writes, ACK polling, and production traceability
- Validated and debugged I²C EEPROM communication using a Saleae logic analyzer, resolving issues involving device addressing, write-cycle timing, page boundaries, and hardware/software integration
- Built a Python BOM availability tool using Playwright and openpyxl to automate component stock checks across approved distributors and generate color-coded sourcing reports highlighting unavailable or insufficient-stock parts

Electrical and Computer Engineering Department, University of Delaware — Newark, DE
Undergraduate Researcher — June 2025 - August 2025

- Implemented open-source pipeline for RTL-GDSII chip design flow, gate-level netlist activity simulation, dynamic power tracing, and side-channel analysis
- Optimized python scripts for simulating dynamic power consumption of cryptographic hardware, improving Pearson correlation coefficients in subsequent correlation power analysis simulations
- Achieved greater efficiency with open-source simulation pipeline flow (16 hours) compared to commercial simulation flows (24 hours)
- Extended analysis beyond gate-level by incorporating layout-level data, enabling more accurate power simulations and side-channel evaluations compared to commercial flows

Phase Sensitive Innovations — Newark, DE
Electrical Engineering Intern — January 2024, June 2024 - August 2024

- Soldered, cleaned, and flattened PCBs for integration into RF antenna arrays
- Wrote and revised technical documents detailing lab equipment operation and standardized procedures
- Operated wire bonders on PCBs and performed integrity testing to ensure reliable connections

PROJECT EXPERIENCE

Semiconductor Solar Cell Fabrication — Newark, DE
Project — March 2026 - May 2026

- Fabricated silicon solar cells through wafer preparation, diffusion, HF etching, and metal contact deposition using electron-beam physical vapor deposition
- Deposited Al and Ti/Pd/Ag thin-film contact and analyzed deposition rate, thickness, and beam-current data
- Characterized fabricated devices using four-point probe measurements, illuminated I-V testing, and quantum efficiency measurements from 340-1200 nm, achieving efficiencies up to 7.3% and corrected QE up to 77%

Designing 3D Printed Electronics using Artificial Intelligence — Newark, DE
Senior Design Project — August 2026 - Present

- Developing an AI-assisted design workflow for 3D-printed electronics to streamline generation of printable electronic geometries from user-defined design requirements

Line-Following Robot based on Basys MX3 — Newark, DE
Project — November 2024 - December 2024

- Developed C-based programs on a microcontroller to enable autonomous line detection and tracking
- Integrated sensors, motors, LEDs, LCDs, RGBs, microphones, timers, and SSD for real-time status display and audio-based control during operation
- Demonstrated competitive success by outperforming opponents in course-end races